

HW Class12 Pt.2 (Population analysis) [Q13 Q14 BoxPlot]

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Population Scale Analysis (HW)

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

The sample sizes are 108 for A/A with median 31.24847, 233 for A/G with median 25.06486, and 121 for G/G with median 20.07363.

```
# Read the file
data <- read.table("https://bioboot.github.io/bggn213_W19/class-material/rs8067378_ENSG000000000000.txt",
                    header = TRUE)

# Preview the data
head(data)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
# Sample size for each genotype
table(data$geno)
```

A/A	A/G	G/G
108	233	121

```
# Median expression levels by genotype group

median_expr <- tapply(data$exp, data$geno, median)

median_expr
```

```
      A/A      A/G      G/G
31.24847 25.06486 20.07363
```

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

The boxplot shows that individuals with the A/A genotype have higher ORMDL3 expression than individuals with the G/G genotype. The A/G group appears intermediate between A/A and G/G. This suggests that the SNP genotype is associated with ORMDL3 expression, meaning the SNP may affect ORMDL3 expression levels. Specifically, the G/G genotype is associated with lower ORMDL3 expression compared with A/A in this plot.

```
library(ggplot2)

ggplot(data, aes(x = geno, y = exp, fill = geno)) +
  geom_boxplot(notch = TRUE, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5, color = "gray30", size = 1.5) +
  scale_fill_manual(values = c("A/A" = "lightcoral",
                              "A/G" = "lightgreen",
                              "G/G" = "lightblue")) +
  labs(x = "Genotype", y = "Expression") +
  theme_gray() +
  theme(legend.position = "none")
```

